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ARTICLE OF FOOTWEAR HAVING A CLOSURE SYSTEM

Abstract

An article of footwear including an upper that is attached to a sole, a longitudinal axis that extends between a rear end and a front end of the article of footwear, a midfoot region that is disposed between a forefoot region and a heel region, a heel component disposed adjacent the rear end within the heel region, a fastening system including a fastening mechanism that is operably engaged with a cord, and a panel. The heel component has a midsection that includes padding. The panel includes an edge that is curved, and the cord is laced through a plurality of eyelets arranged on the panel. The cord extends on a lateral side of the upper so that a first segment extends diagonally downward from a top section toward the sole before extending diagonally upwardly toward the top section, and the first segment crosses over a second segment of the cord.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] This application is a continuation of U.S. patent application Ser. No. 18/085,108, filed on Dec. 20, 2022, which claims the benefit of and priority to U.S. Provisional Application No. 63/292,289, filed on Dec. 21, 2021, which are both incorporated by reference herein in their entirety.

REFERENCE REGARDING FEDERALLY SPONSORED RESEARCH OR DEVELOPMENT

[0002] Not applicable

SEQUENCE LISTING

[0003] Not applicable

BACKGROUND

1. Field of the Invention

[0004] The present disclosure relates generally to an article of footwear including a fastening system or closure system.

2. Description of the Background

[0005] Many conventional shoes or other articles of footwear generally comprise an upper and a sole attached to a lower end of the upper. Conventional shoes further include an internal space, i.e., a void or cavity, which is created by interior surfaces of the upper and sole, that receives a foot of a user before securing the shoe to the foot. The sole is attached to a lower surface or boundary of the upper and is positioned between the upper and the ground. As a result, the sole typically provides stability and cushioning to the user when the shoe is being worn. In some instances, the sole may include multiple components, such as an outsole, a midsole, and a top portion. The outsole may provide traction to a bottom surface of the sole, and the midsole may be attached to an inner surface of the outsole, and may provide cushioning or added stability to the sole. For example, a sole may include a particular foam material that may increase stability at one or more desired locations along the sole, or a foam material that may reduce stress or impact energy on the foot or leg when a user is running, walking, or engaged in another activity. The sole may also include additional components, such as plates, embedded with the sole to increase the overall stiffness of the sole and reduce energy loss during use.

[0006] The upper generally extends upward from the sole and defines an interior cavity that completely or partially encases a foot. In most cases, the upper extends over the instep and toe regions of the foot, and across medial and lateral sides thereof. Many articles of footwear may also include a tongue that extends across the instep region to bridge a gap between edges of medial and lateral sides of the upper, which define an opening into the cavity. The tongue may also be disposed below a lacing system and between medial and lateral sides of the upper, to allow for adjustment of shoe tightness. The tongue may further be manipulatable by a user to permit entry or exit of a foot from the internal space or cavity. In addition, the lacing system may allow a user to adjust certain dimensions of the upper or the sole, thereby allowing the upper to accommodate a wide variety of foot types having varying sizes and shapes.

[0007] The upper of many shoes may comprise a wide variety of materials, which may be utilized to form the upper and chosen for use based on one or more intended uses of the shoe. The upper may also include portions comprising varying materials specific to a particular area of the upper. For example, added stability may be desirable at a front of the upper or adjacent a heel region so as to provide a higher degree of resistance or rigidity. In contrast, other portions of a shoe may include

a soft woven textile to provide an area with stretch-resistance, flexibility, air-permeability, or moisture-wicking properties.

[0008] However, in many cases, articles of footwear having uppers with an increased comfort and better fit are desired, along with improved closure mechanisms.

SUMMARY

[0009] An article of footwear, as described herein, may have various configurations. The article of footwear may have an upper and a sole structure connected to the upper.

[0010] In some aspects, a closure system for an article of footwear includes an actuation mechanism, a release mechanism, a closure mechanism, and a cord. The cord is configured to be operably engaged with an upper of the footwear. The actuation mechanism is configured to be pulled to actuate the closure system to adjust the footwear from a loosened configuration to a tightened configuration. In addition, the release mechanism is configured to adjust the footwear from the tightened configuration to the loosened configuration.

[0011] In some embodiments, the actuation mechanism is a strap and the release mechanism is a dial. In some embodiments, the closure mechanism is a disc. Further, the closure mechanism is configured to hold the footwear in the tightened configuration. The cord is comprised of multiple segments operably coupled to one another by the closure mechanism. In some embodiments, the tightened configuration includes varying levels of tightness, which include a first level, a second level, and a third level. Additionally, the third level is tighter than the second level and the second level is tighter than the first level.

[0012] In another aspect, a closure system is provided for an article of footwear having a sole connected to an upper. The closure system includes an actuation mechanism, a release mechanism, and a closure mechanism. Further, the closure system is decentralized and is operably connected by a cord that extends along the footwear.

[0013] In some embodiments, the actuation mechanism is arranged in a heel region of the footwear and the release mechanism is arranged in a midfoot region of the footwear. In some embodiments, the closure mechanism is positioned on the upper within a midfoot region of the footwear. In other embodiments, the closure mechanism is disposed within the sole of the footwear. In some embodiments, the closure mechanism and the release mechanism are arranged in a midfoot region of the footwear. Further, the cord includes a first segment and a second segment. The first segment extends between the actuation mechanism and the closure mechanism and the second segment extends from the closure mechanism to a retention member. Additionally, the first segment of the cord is formed of an elastic material that is configured to retract the actuation mechanism from an actuation configuration to a rest configuration.

[0014] In still another aspect, a centralized closure system is provided for an article of footwear. The centralized closure system includes an actuation mechanism, a release mechanism, and a closure mechanism. The centralized closure system is configured to provide haptic feedback to a user. Further, the actuation mechanism is configured to be pulled to actuate the closure system, and the actuation mechanism is configured to be retracted after actuation of the closure system.

[0015] In some embodiments, the closure mechanism is operably engaged with a cord to tighten the footwear. Further, the actuation mechanism is configured to be pulled away from the closure mechanism to actuate the closure system. The actuation mechanism is operably engaged with a coil spring disposed within the closure mechanism, with the coil spring being configured to retract the actuation mechanism toward the closure mechanism. In some embodiments, the actuation mechanism and the closure mechanism are operably coupled to one another and disposed within a midfoot region of the footwear.

[0016] In some aspects, an article of footwear includes an upper that is attached to a sole, a longitudinal axis that extends between a rear end and a front end of the article of footwear, and a midfoot region that is disposed between a forefoot region and a heel region. The article of footwear further comprises a heel component disposed adjacent the rear end within the heel region. The heel

component has a midsection that includes padding or foam. Further, the article of footwear includes a fastening system having a release mechanism that is operably engaged with a cord. The release mechanism includes a disc-shaped housing that is mounted to a top section of the upper within a midfoot region. The cord is laced through a plurality of conduits arranged on the upper. A distance is defined between the midsection of the heel component and the upper. The distance is configured to be adjusted between a first configuration and a second configuration. The midsection of the heel component is configured to be translated rearward and upward relative to the longitudinal axis, and the midsection of the heel component is configured to be translated forward and downward relative to the longitudinal axis.

[0017] In some embodiments, the midsection of the heel component is configured to be gripped by a user. In some embodiments, the midsection extends between a first end and a second end of the heel component. The first end is arranged adjacent the medial side and the second end is arranged adjacent the lateral side. In some embodiments, the midsection of the heel component is concavely curved between the medial side and the lateral side relative to a toe end of the article of footwear. In some embodiments, the midsection of the heel component is configured to at least partially deform when translated between the first configuration and the second configuration. In some embodiments, at least a portion of the cord is arranged on the upper between a medial side and a lateral side of the article of footwear to form a V-shaped pattern. In some embodiments, the release mechanism is disposed centrally between the lateral side and the medial side of the article of footwear.

[0018] Other aspects of the article of footwear, including features and advantages thereof, will become apparent to one of ordinary skill in the art upon examination of the figures and detailed description herein. Therefore, all such aspects of the article of footwear are intended to be included in the detailed description and this summary.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0019] FIG. 1 is a perspective view of a bottom and medial side of an article of footwear configured as a right shoe that includes an upper and a sole structure, according to an embodiment of the disclosure;

[0020] FIG. 2 is a top view of the article of footwear of FIG. 1;

[0021] FIG. 3 is a top plan view of the article of footwear of FIG. 1, with an upper removed and a user's skeletal foot structure overlaid thereon;

[0022] FIG. 4 is a schematic representation of a perspective view of a medial side and a heel end of an article of footwear configured as a left shoe, according to an embodiment of the disclosure; and

[0023] FIG. 5 is a schematic representation of a lateral side of the article of footwear of FIG. 4 in an as-worn and actuated configuration;

[0024] FIG. 6 is a schematic representation of a top plan view of the article of footwear of FIG. 4 in an unworn and rest configuration;

[0025] FIG. 7 is a schematic representation of a perspective lateral side and toe end view of another article of footwear configured as a left shoe, according to an embodiment of the disclosure;

[0026] FIG. 8 is a schematic representation of a perspective lateral side and toe end view of yet another article of footwear configured as a left shoe, according to an embodiment of the disclosure;

[0027] FIG. 9 is a schematic representation of an exploded, partial view of a closure system for the article of footwear of FIG. 8;

[0028] FIG. 10 is a schematic representation of a perspective lateral side and toe end view of still another article of footwear configured as a left shoe, according to an embodiment of the disclosure;

[0029] FIG. 11 is a schematic representation of a perspective bottom and medial side view of yet

another article of footwear configured as a right shoe, according to an embodiment of the disclosure; and

[0030] FIG. 12 is a schematic representation of a top plan view of still another article of footwear configured as a left shoe, with an upper removed, according to an embodiment of the disclosure.

DETAILED DESCRIPTION OF THE DRAWINGS

[0031] The following discussion and accompanying figures disclose various embodiments or configurations of a shoe and a sole structure. Although embodiments of a shoe or sole structure are disclosed with reference to a sports shoe, such as a running shoe, tennis shoe, basketball shoe, etc., concepts associated with embodiments of the shoe or the sole structure may be applied to a wide range of footwear and footwear styles, including cross-training shoes, football shoes, golf shoes, hiking shoes, hiking boots, ski and snowboard boots, soccer shoes and cleats, walking shoes, and track cleats, for example. Concepts of the shoe or the sole structure may also be applied to articles of footwear that are considered non-athletic, including dress shoes, sandals, loafers, slippers, and heels. In addition to footwear, particular concepts described herein may also be applied and incorporated in other types of apparel or other athletic equipment, including helmets, padding or protective pads, shin guards, and gloves. Even further, particular concepts described herein may be incorporated in cushions, backpack straps, golf clubs, or other consumer or industrial products. Accordingly, concepts described herein may be utilized in a variety of products.

[0032] The term “about,” as used herein, refers to variation in the numerical quantity that may occur, for example, through typical measuring and manufacturing procedures used for articles of footwear or other articles of manufacture that may include embodiments of the disclosure herein; through inadvertent error in these procedures; through differences in the manufacture, source, or purity of the ingredients used to make the compositions or mixtures or carry out the methods; and the like. Throughout the disclosure, the terms “about” and “approximately” refer to a range of values $\pm 5\%$ of the numeric value that the term precedes.

[0033] The present disclosure is directed to an article of footwear and/or specific components of the article of footwear, such as an upper and/or a sole or sole structure. The upper may comprise a knitted component, a woven textile, and/or a non-woven textile. The knitted component may be made by knitting of yarn, the woven textile by weaving of yarn, and the non-woven textile by manufacture of a unitary non-woven web. Knitted textiles include textiles formed by way of warp knitting, weft knitting, flat knitting, circular knitting, and/or other suitable knitting operations. The knit textile may have a plain knit structure, a mesh knit structure, and/or a rib knit structure, for example. Woven textiles include, but are not limited to, textiles formed by way of any of the numerous weave forms, such as plain weave, twill weave, satin weave, dobbin weave, jacquard weave, double weaves, and/or double cloth weaves, for example. Non-woven textiles include textiles made by air-laid and/or spun-laid methods, for example. The upper may comprise a variety of materials, such as a first yarn, a second yarn, and/or a third yarn, which may have varying properties or varying visual characteristics.

[0034] FIGS. 1-3 depict an exemplary embodiment of an article of footwear **100** including an upper **102** and a sole structure **104**. The upper **102** is attached to the sole structure **104** and together define an interior cavity **106** (see FIGS. 2 and 3) into which a foot may be inserted. For reference, the article of footwear **100** defines a forefoot region **108**, a midfoot region **110**, and a heel region **112** (see FIGS. 2 and 3). The forefoot region **108** generally corresponds with portions of the article of footwear **100** that encase portions of the foot that includes the toes, the ball of the foot, and joints connecting the metatarsals with the toes or phalanges. The midfoot region **110** is proximate and adjoining the forefoot region **108**, and generally corresponds with portions of the article of footwear **100** that encase the arch of the foot, along with the bridge of the foot. The heel region **112** is proximate and adjoining the midfoot region **110** and generally corresponds with portions of the article of footwear **100** that encase rear portions of the foot, including the heel or calcaneus bone, the ankle, and/or the Achilles tendon.

[0035] Many conventional footwear uppers are formed from multiple elements, e.g., textiles, polymer foam, polymer sheets, leather, and synthetic leather, which are joined through bonding or stitching at a seam. In some embodiments, the upper **102** of the article of footwear **100** is formed from a knitted structure or knitted components. In various embodiments, a knitted component may incorporate various types of yarn that may provide different properties to an upper. For example, one area of the upper **102** may be formed from a first type of yarn that imparts a first set of properties, and another area of the upper **102** may be formed from a second type of yarn that imparts a second set of properties. Using this configuration, properties of the upper **102** may vary throughout the upper **102** by selecting specific yarns for different areas of the upper **102**.

[0036] With reference to the material(s) that comprise the upper **102**, the specific properties that a particular type of yarn will impart to an area of a knitted component may at least partially depend upon the materials that form the various filaments and fibers of the yarn. For example, cotton may provide a soft effect, biodegradability, or a natural aesthetic to a knitted material. Elastane and stretch polyester may each provide a knitted component with a desired elasticity and recovery. Rayon may provide a high luster and moisture absorbent material, wool may provide a material with an increased moisture absorbance, nylon may be a durable material that is abrasion-resistant, and polyester may provide a hydrophobic, durable material.

[0037] Other aspects of a knitted component may also be varied to affect the properties of the knitted component and provide desired attributes. For example, a yarn forming a knitted component may include monofilament yarn or multifilament yarn, or the yarn may include filaments that are each formed of two or more different materials. In addition, a knitted component may be formed using a particular knitting process to impart an area of a knitted component with particular properties. Accordingly, both the materials forming the yarn and other aspects of the yarn may be selected to impart a variety of properties to particular areas of the upper **102**.

[0038] In some embodiments, an elasticity of a knit structure may be measured based on comparing a width or length of the knit structure in a first, non-stretched state to a width or length of the knit structure in a second, stretched state after the knit structure has a force applied to the knit structure in a lateral direction. In further embodiments, the upper **102** may also include additional structural elements. For example, in some embodiments, a heel plate or cover (not shown) may be provided on the heel region **112** to provide added support to a heel of a user. In some instances, other elements, e.g., plastic material, logos, trademarks, etc., may also be applied and fixed to an exterior surface using glue or a thermoforming process. In some embodiments, the properties associated with the upper **102**, e.g., a stitch type, a yarn type, or characteristics associated with different stitch types or yarn types, such as elasticity, aesthetic appearance, thickness, air permeability, or scuff-resistance, may be varied.

[0039] Referring again to FIG. **1**, the sole structure **104** is connected or secured to the upper **102** and extends between a foot of a user and the ground when the article of footwear **100** is worn by the user. The sole structure **104** may include one or more components, which may include an outsole, a midsole, a heel, a vamp, and/or an insole. For example, in some embodiments, a sole structure may include an outsole that provides structural integrity to the sole structure, along with providing traction for a user, a midsole that provides a cushioning system, and an insole that provides support for an arch of a user. In addition, the insole may be a strobel board, a forefoot board, a lasting board, etc., or a combination thereof, and the insole may be provided between the upper **102** and the sole structure **104**, or the insole may be provided as part of the upper **102**.

[0040] Furthermore, the insole can be positioned within the interior cavity **106** of the upper **102**, which can be in direct contact with a user's foot while an article of footwear **100** is being worn. Moreover, the upper **102** may also include a liner (not shown) that can increase comfort, for example, by reducing friction between the foot of the user and the upper **102**, the sole **104**, the insole, or the like, and/or by providing moisture wicking properties. The liner may line the entirety of the interior cavity **106** or only a portion thereof. In some embodiments, a binding (not shown)

may surround an opening of the interior cavity **106** to secure the liner to the upper **102** and/or to provide an aesthetic element on the article of footwear **100**.

[0041] Referring to FIGS. **2** and **3**, the article of footwear **100** also defines a lateral side **116** and a medial side **118**. When a user is wearing the shoes, the lateral side **116** corresponds with an outside-facing portion of the article of footwear **100** while the medial side **118** corresponds with an inside-facing portion of the article of footwear **100**. As such, the article of footwear **100** has opposing lateral sides **116** and medial sides **118**. The medial side **118** and the lateral side **116** adjoin one another along a longitudinal central plane or axis **120** of the article of footwear **100**, which is coplanar with the longitudinal axis L of FIG. **1**. As will be further discussed herein, the longitudinal central plane or axis **120** may demarcate a central, intermediate axis between the medial side **118** and the lateral side **116** of the article of footwear **100**. Put differently, the longitudinal plane or axis **120** may extend between a rear, proximal end **122** of the article of footwear **100** and a front, distal end **124** of the article of footwear **100** and may continuously define a middle of an insole **126**, the sole structure **104**, and/or the upper **102** of the article of footwear **100**, i.e., the longitudinal plane or axis **120** is a straight axis extending through the rear, proximal end **122** of the heel region **112** to the front, distal end **124** of the forefoot region **108**.

[0042] Unless otherwise specified, and referring to FIGS. **2** and **3**, the article of footwear **100** may be defined by the forefoot region **108**, the midfoot region **110**, and the heel region **112**. The forefoot region **108** may generally correspond with portions of the article of footwear **100** that encase portions of a foot **128** that include the toes or phalanges **130**, the ball of the foot **132**, and one or more of the joints **134** that connect the metatarsals **136** of the foot **128** with the toes or phalanges **130**. The midfoot region **110** is proximate and adjoins the forefoot region **108**. The midfoot region **110** generally corresponds with portions of the article of footwear **100** that encase an arch of a foot **128**, along with a bridge of the foot **128**. The heel region **112** is proximate to the midfoot region **110** and adjoins the midfoot region **110**. The heel region **112** generally corresponds with portions of the article of footwear **100** that encase rear portions of the foot **128**, including the heel or calcaneus bone **138**, the ankle (not shown), and/or the Achilles tendon (not shown).

[0043] Still referring to FIGS. **2** and **3**, the forefoot region **108**, the midfoot region **110**, the heel region **112**, the medial side **118**, and the lateral side **116** are intended to define boundaries or areas of the article of footwear **100**. To that end, the forefoot region **108**, the midfoot region **110**, the heel region **112**, the medial side **118**, and the lateral side **116** generally characterize sections of the article of footwear **100**. Certain aspects of the disclosure may refer to portions or elements that are coextensive with one or more of the forefoot region **108**, the midfoot region **110**, the heel region **112**, the medial side **118**, and/or the lateral side **116**. Further, both the upper **102** and the sole structure **104** may be characterized as having portions within the forefoot region **108**, the midfoot region **110**, the heel region **112**, and/or along the medial side **118** and/or the lateral side **116**. Therefore, the upper **102** and the sole structure **104**, and/or individual portions of the upper **102** and the sole structure **104**, may include portions thereof that are disposed within the forefoot region **108**, the midfoot region **110**, the heel region **112**, and/or along the medial side **118** and/or the lateral side **116**.

[0044] Still referring to FIGS. **2** and **3**, the forefoot region **108**, the midfoot region **110**, the heel region **112**, the medial side **118**, and the lateral side **116** are shown in detail. The forefoot region **108** extends from a toe end **140** to a widest portion **142** of the article of footwear **100**. The widest portion **142** is defined or measured along a first line **144** that is perpendicular with respect to the longitudinal axis **120** that extends from a distal portion of the toe end **140** to a distal portion of a heel end **146**, which is opposite the toe end **140**. The midfoot region **110** extends from the widest portion **142** to a thinnest portion **148** of the article of footwear **100**. The thinnest portion **148** of the article of footwear **100** is defined as the thinnest portion of the article of footwear **100** measured across a second line **150** that is perpendicular with respect to the longitudinal axis **120**. The heel region **112** extends from the thinnest portion **148** to the heel end **146** of the article of footwear **100**.

[0045] It should be understood that numerous modifications may be apparent to those skilled in the art in view of the foregoing description, and individual components thereof, may be incorporated into numerous articles of footwear. Accordingly, aspects of the article of footwear **100** and components thereof, may be described with reference to general areas or portions of the article of footwear **100**, with an understanding the boundaries of the forefoot region **108**, the midfoot region **110**, the heel region **112**, the medial side **118**, and/or the lateral side **116** as described herein may vary between articles of footwear. However, aspects of the article of footwear **100** and individual components thereof, may also be described with reference to exact areas or portions of the article of footwear **100** and the scope of the appended claims herein may incorporate the limitations associated with these boundaries of the forefoot region **108**, the midfoot region **110**, the heel region **112**, the medial side **118**, and/or the lateral side **116** discussed herein.

[0046] Still referring to FIGS. 2 and 3, the medial side **118** begins at the distal, toe end **140** and bows outward along an inner side of the article of footwear **100** along the forefoot region **108** toward the midfoot region **110**. The medial side **118** reaches the first line **144**, at which point the medial side **118** bows inward, toward the central, longitudinal axis **120**. The medial side **118** extends from the first line **144**, i.e., the widest portion **142**, toward the second line **150**, i.e., the thinnest portion **148**, at which point the medial side **118** enters into the midfoot region **110**, i.e., upon crossing the first line **144**. Once reaching the second line **150**, the medial side **118** bows outward, away from the longitudinal, central axis **120**, at which point the medial side **118** extends into the heel region **112**, i.e., upon crossing the second line **150**. The medial side **118** then bows outward and then inward toward the heel end **146**, and terminates at a point where the medial side **118** meets the longitudinal, central axis **120**.

[0047] The lateral side **116** also begins at the distal, toe end **140** and bows outward along an outer side of the article of footwear **100** along the forefoot region **108** toward the midfoot region **110**. The lateral side **116** reaches the first line **144**, at which point the lateral side **116** bows inward, toward the longitudinal, central axis **120**. The lateral side **116** extends from the first line **144**, i.e., the widest portion **142**, toward the second line **150**, i.e., the thinnest portion **148**, at which point the lateral side **116** enters into the midfoot region **110**, i.e., upon crossing the first line **144**. Once reaching the second line **150**, the lateral side **116** bows outward, away from the longitudinal, central axis **120**, at which point the lateral side **116** extends into the heel region **112**, i.e., upon crossing the second line **150**. The lateral side **116** then bows outward and then inward toward the heel end **146**, and terminates at a point where the lateral side **116** meets the longitudinal, central axis **120**.

[0048] Still referring to FIGS. 2 and 3, the upper **102** extends along the lateral side **116** and the medial side **118**, and across the forefoot region **108**, the midfoot region **110**, and the heel region **112** to house and enclose a foot of a user. When fully assembled, the upper **102** also includes an interior surface **162** and an exterior surface **164**. The interior surface **162** faces inward and generally defines the interior cavity **106**, and the exterior surface **164** of the upper **102** faces outward and generally defines an outer perimeter or boundary of the upper **102**. The upper **102** also includes an opening **166** that is at least partially located in the heel region **112** of the article of footwear **100**, which provides access to the interior cavity **106** and through which a foot may be inserted and removed. In some embodiments, the upper **102** may also include an instep region **168** that extends from the opening **166** in the heel region **112** over an area corresponding to an instep of a foot to an area proximate the forefoot region **108**. The instep region **168** may comprise an area similar to where a tongue **170** of the present embodiment is disposed. In some embodiments, the upper **102** does not include the tongue **170**, i.e., the upper **102** is tongueless.

[0049] In the illustrated embodiment, the sole structure **104** includes a midsole **172** and an outsole **174**. The outsole **174** may define a bottom end or bottom surface **176** of the sole structure **104** across the heel region **112**, the midfoot region **110**, and the forefoot region **108**. Further, the outsole **174** may be a ground-engaging portion or include a ground-engaging surface of the sole structure **104** and may be opposite of the insole thereof. As illustrated in FIG. 1, the bottom surface **176** of

the outsole **174** may include a tread pattern **178** that can include a variety of shapes and configurations. The outsole **174** may be formed from one or more materials to impart durability, wear-resistance, abrasion resistance, or traction to the sole structure **104**. In some embodiments, the outsole **174** may be formed from any kind of elastomer material, e.g., rubber, including thermoset elastomers or thermoplastic elastomers, or a thermoplastic material, e.g., thermoplastic polyurethane (TPU). In some embodiments, the outsole **174** may define a shore A hardness up to 95. In addition, the outsole **174** may be manufactured by a process involving injection molding, vulcanization, printing layer by layer, i.e., additive manufacturing systems or methods, and the like. [0050] Still referring to FIG. **1**, the midsole **172** may be individually constructed from a thermoplastic material, such as polyurethane (PU), for example, and/or an ethylene-vinyl acetate (EVA), copolymers thereof, or a similar type of material. In other embodiments, the midsole **172** may be an EVA-Solid-Sponge (“ESS”) material, an EVA foam (e.g., PUMA® ProFoam Lite™, IGNITE Foam), polyurethane, polyether, an olefin block copolymer, organosheets, a thermoplastic material (e.g., a thermoplastic polyurethane, a thermoplastic elastomer, a thermoplastic polyolefin, etc.), or a supercritical foam. The midsole **172** may be a single polymeric material or may be a blend of materials, such as an EVA copolymer, a thermoplastic polyurethane, a polyether block amide (PEBA) copolymer, and/or an olefin block copolymer. One example of a PEBA material is PEBAX®. In some embodiments, the midsole **172** is manufactured by a process involving injection molding, vulcanization, printing layer by layer, i.e., additive manufacturing systems or methods, and the like.

[0051] In embodiments where the midsole **172** is formed from a supercritical foaming process, the supercritical foam may comprise micropore foams or particle foams, such as a TPU, EVA, PEBAX®, or mixtures thereof, manufactured using a process that is performed within an autoclave, an injection molding apparatus, or any sufficiently heated/pressurized container that can process the mixing of a supercritical fluid (e.g., CO.sub.2, N.sub.2, or mixtures thereof) with a material (e.g., TPU, EVA, polyolefin elastomer, or mixtures thereof) that is preferably molten. During an exemplary process, a solution of supercritical fluid and molten material is pumped into a pressurized container, after which the pressure within the container is released, such that the molecules of the supercritical fluid rapidly convert to gas to form small pockets within the material and cause the material to expand into a foam. In further embodiments, the midsole **172** may be formed using alternative methods known in the art, including the use of an expansion press, an injection machine, a pellet expansion process, a cold foaming process, a compression molding technique, die cutting, or any combination thereof. For example, the midsole **172** may be formed using a process that involves an initial foaming step in which supercritical gas is used to foam a material and then compression molded or die cut to a particular shape.

[0052] FIG. **4** depicts a schematic representation of an article of footwear **200** having a fastening system or closure system **204** that includes an actuation mechanism in the form of a strap or heel component **208** that is operably engaged with a cord **212** to actuate the closure system **204** to tighten the article of footwear **200** around a user's foot, e.g., adjusting the footwear **200** from a loosened configuration to a tightened configuration. Further, the closure system **204** includes a release mechanism **216** in the form of a switch or dial that is operably engaged with the cord **212** for unlocking and/or releasing the footwear **200** after being tightened, e.g., from the tightened configuration to the loosened configuration. In addition, the closure system **204** includes a closure mechanism or a disc **220** that is operably engaged with the cord **212** to selectively lock or hold the footwear **200** in a tightened configuration and to selectively be unlocked or released by operation of the release mechanism **216**.

[0053] Referring to FIG. **4**, the strap **208** is arranged on the upper **102** and within the heel region **112** of the footwear **200**. In particular, the strap **208** is located at the heel end **146** of the footwear **200** and extends in a generally U-shaped path between a pair of apertures **224a**, **224b**, only one of which is visible from the view of FIG. **4**. The apertures **224a**, **224b** are disposed on the medial side

118 and the lateral side **116**, respectively, of the upper **102** within the heel region **112**. In the illustrated embodiment, the strap **208** is spaced apart from the sole **104** and below a collar **228** that surrounds the opening **166** to the interior cavity **106** of the upper **102**. Further, the apertures **224a**, **224b** are arranged an equal distance from the sole **104** and also spaced apart from one another. The apertures **224a**, **224b** are connected to channels (not shown) formed within the upper **102** and extending in the heel region **112** toward the midfoot region **110**.

[0054] In FIG. 4, the strap **208** is depicted in a rest configuration in which a gap **232** is formed between the upper **102** and the strap **208**. In particular, in the rest configuration the midsection **236** of the strap **208** is located a distance **G** from the upper **102**, such that the strap **208** extends from the heel end **146** of the upper **102** in a rearward direction and an upward direction relative to the longitudinal axis **L**. The midsection **236** of the strap **208** is relatively thicker than the surrounding portions of the strap **208**. To that end, the midsection **236** may include padding, e.g., foam, to provide comfort, durability, and resilience when gripped by a user. Further, the midsection **236** extends between a first end **240** and a second end **244** of the strap **208**, such that the strap **208** and the midsection **236** thereof curves concavely between the lateral side **116** and the medial side **118** relative to the toe end **140**. In the rest configuration, the first and second ends **240**, **244** of the strap **208** are disposed within or adjacent the apertures **224a**, **224b** and the channels (not shown) of the upper **102**. It can be appreciated from FIG. 4 that the first and second ends **240**, **244** of the strap **208** are coupled to the cord **212** for operably engagement and/or actuation of the closure system **204**. In the rest configuration, the first and second ends **240**, **244** are coupled to portions of the cord **212** adjacent the apertures **224a**, **224b** and at least partially concealed within the upper **102**. It is contemplated that the strap **208** is formed of a resilient material that is configured to retain its shape in the rest configuration, as illustrated in FIG. 4.

[0055] Still referring to FIG. 4, the cord **212** is arranged on the upper **102** within the heel region **112** and the midfoot region **110** of the footwear **200**. In the illustrated embodiment, the cord **212** defines a winding region **248** that may extend across the upper **102** and/or the sole **104** and across several regions of the footwear **200**. The cord **212** is comprised of a plurality of segments, including a first segment **252** that extends between the strap **208** and the release mechanism **216** and/or disc **220**. Further, the cord **212** includes a second segment **256** extending between the release mechanism **216** and/or disc **220** and a retention member **260** arranged on the upper **102**. The retention member **260** may be provided in the form of a loop or series of loops, although other configurations are possible. In the illustrated embodiment, the retention member **260** is located within the forefoot region **108** of the upper **102** and spaced apart from the release mechanism **216** and/or disc **220**, although other configurations are possible. The second segment **256** of the cord **212** may be coupled to the sole **104** at a point between the retention member **260** and the release mechanism **216** and/or disc **220**. In some embodiments, a plurality of retention members **260** are arranged on the upper **102** and/or the sole **104** for retaining the cord **212**.

[0056] The cord **212**, including the first segment **252** and the second segment **256**, is configured to translate tensional forces applied by the strap **208** and/or the disc **220** into compression, e.g., tightening, forces and to distribute such compression forces about the footwear **200** to tighten around a user's foot. In the illustrated embodiment, the first segment **252** and the second segment **256** of the cord **212** extend along both the lateral side **116** and the medial side **118** of the upper **102**, although other configurations are possible. In particular, the first segment **252** of the cord **212** of the closure system **204** is disposed in a V-shape pattern between apertures **224a**, **224b** and the disc **220** and/or release mechanism **216**, where a first lowermost point **264** is positioned on the upper **102** adjacent the sole **104** within the heel region **112**. It is contemplated that the first lowermost point **264** may be disposed on or within the sole **104** and within a different region of the footwear **200**. The second segment **256** of the cord **212** may also be disposed in a V-shape pattern between the disc **220** and/or release mechanism **216** and the retention member **260**, where a second lowermost point **268** is positioned on the upper **102** adjacent the sole **104** within the midfoot region

110. It is contemplated that the second lowermost point **268** of the second segment **256** may be disposed on or within the sole **104** and within a different region of the footwear **200**. Further, it is contemplated that the first lowermost point **264** of the first segment **252** and the second lowermost point **268** of the second segment **256** can be positioned at different elevations relative to the sole **104** and/or to the longitudinal axis **L**.

[0057] It is contemplated that the cord **212** includes multiple, distinct sections joined or coupled together, such as the first segment **252** being jointed to the second segment **256** by the release mechanism **216** and/or the disc **220**. In some embodiments, the cord **212** is comprised of multiple sections that are integrally formed as a unitary member. Further, the cord **212** may be formed of a variety of resilient and flexible materials, such as, e.g., elastic or thermoplastic materials, woven or non-woven fabric or fibrous materials, or some combination thereof. In some embodiments, the cord **212** is fastened to the upper **102** and/or the sole **104** by a plurality of the retention members **260** arranged in several regions of the footwear **200**. Additionally or alternatively, the cord **212** may be laced or threaded through a plurality of conduits **270** formed within the upper **102** and/or the sole **104**, such as, e.g., a passage formed between layers or substrates comprising the upper **102** and/or the sole **104**. In some embodiments, the conduits **270** may extend continuously along the upper **102** and/or the sole **104** such that the entire cord **212** is enclosed therein. In some embodiments, the conduit **270** is provided in distinct sections such that portions or segments of the cord **212** are exposed outside of the upper **102** or inside the interior cavity **106**.

[0058] Further, the cord **212** may be attached to the insole **126** along a periphery or portions thereof. It is contemplated that the cord **212** can be woven and/or weaved into the upper **102**, or layers or substrates thereof, such that the upper **102** is formed integrally with the cord **212** embedded therein. In this way, manufacturing of the upper **102** and the cord **212** can be accomplished efficiently in fewer steps and with less labor. Further, it is contemplated that the cord **212** is fashioned from multiple materials having different material properties for achieving different functions. For example, segments of the cord **212** may be configured to be more flexible than other segments to control the tensional forces distributed there along in particular regions or areas of the footwear **200**. Further, the cord **212** is configured, via material properties and attachment or integration with the upper **102**, to withstand repeated tension and flexure throughout repeated use, as well as to withstand outdoor environments.

[0059] Referring to FIG. **4**, the release mechanism **216** is arranged on a top section **272** of the upper **102** within the midfoot region **110** of the footwear **200**. In the illustrated embodiment, the release mechanism **216** is provided as a generally rounded component located centrally between the lateral side **116** and the medial side **118** of the footwear **200** and on the tongue **170** of the upper **102** adjacent the opening **166**, although other configurations are possible. The upper **102** of the footwear **200** includes the top section **272** extending from the forefoot region **108** to the heel region **112** and around the opening **166** into the interior cavity **106**. In the illustrated embodiments, the top section **272** is thickened relative to the remainder of the upper **102**. It is contemplated that the top section **272** can include or surround the tongue **170**. The release mechanism **216** is exposed on the upper **102** and is configured to be coupled to the cord **212** between the first segment **252** and the second segment **256**. In this way, the release mechanism **216** is configured to be actuated by a user to operate the closure system **204**. In some embodiments, the release mechanism **216** is configured to be twisted or rotated by a user. In other embodiments, the release mechanism **216** is configured to be pushed or pressed into the upper **102** by a user. In some embodiments, the release mechanism **216** can be twisted and pressed, such that a user can actuate the release mechanism **216** in either manner. Upon actuation, the release mechanism **216** operates a release function of the closure system **204** to allow the footwear **200** to become loosened.

[0060] As illustrated in FIG. **4**, the disc **220** is arranged on the upper **102** within the midfoot region **110** of the footwear **200**. In the illustrated embodiment, the disc **220** is provided as a generally rounded component located centrally between the lateral side **116** and the medial side **118** of the

footwear **200** and on the tongue **170** of the upper **102** adjacent the opening **166**, although other configurations are possible. A portion of the disc **220** is partially exposed on the upper **102** and below the release mechanism **216**, such that the disc **220** is operably coupled to and concealed by the release mechanism **216**. In this way, the disc **220** is configured to be actuated by a user to operate the closure system **204**. Further, the disc **220** is configured to be operably coupled to the cord **212**. The disc **220** is configured to be actuated by the first segment **252** of the cord **212** to tighten and/or loosen the footwear **200**. In addition, the disc **220** is configured to be actuated by the release mechanism **216** to loosen the footwear **200**.

[0061] FIG. 5 depicts a schematic representation of the lateral side **116** of the footwear **200** with the closure system **204**. In the illustrated embodiment, and similar to the medial side **118**, the cord **212** of the closure system **204** includes the first segment **252** extending in a V-shape pattern between the apertures **224a**, **224b** and the disc **220** and/or release mechanism **216**, and the second segment **256** is arranged in a V-shape pattern between the disc **220** and/or release mechanism **216** and the retention member **260**. Further, the first lowermost point **264** of the first segment **252** is arranged in the heel region **112** on the lateral side **116** and the second lowermost point **268** of the second segment **256** is arranged in the forefoot region **108** on the lateral side **116**.

[0062] FIG. 5 depicts the footwear **200** as-worn by a user and with the closure system **204** in an actuation configuration where the strap **208** is pulled or translated rearward and upward relative to the longitudinal axis L (see FIG. 4), away from the apertures **224a**, **224b** on the heel end **146** of the upper **102** by a user's hand **276**. As a result, the gap **232** between the strap **208** and the upper **102** increases in size, particularly with respect to the distance G between the midsection **236** of the strap **208** and the upper **102**, and the midsection **236** of the strap **208** at least partially deforms and/or changes in curvature relative to the toe end **140**. In the actuation configuration depicted in FIG. 5, a portion of the first segment **252** of the cord **212** is pulled through the apertures **224a**, **224b** and out of the upper **102** to become at least partially exposed. To that end, when a user pulls the strap **208** from the rest configuration (see FIG. 4) to the actuation configuration (see FIG. 5), the strap **208** exerts a tensional force that is transmitted through the first segment **252** of the cord **212** to the disc **220** and/or to the second segment **256** of the cord **212**. As a result of the tensional force applied to the cord **212** and transmitted to the disc **220**, the closure system **204** adjusts the footwear **200** to a tightened configuration. As such, the disc **220** and/or the first segment **252** of the cord **212** applies the tensional force to the second segment **256** of the cord **212**, which further distributes the compression force about the footwear **200** in several directions and several regions, allowing for comfortable and consistent tightening of the footwear **200** around a user's foot. As a result of the tensional forces applied to the first segment **252** and the second segment of the cord **212**, the first lowermost point **264** of the first segment **252** may change position, such as, e.g., becoming displaced farther from the sole **104** and the second lowermost point **268** of the second segment **256** may also change position, such as, e.g., becoming displaced from the sole **104**. In this way, the cord **212** is restricted and/or tightened to enclose and compress the user's foot within the interior cavity **106** of the upper **102**. Further, the cord **212** may apply the tensional forces to the lateral side **116** and medial side **118** of the sole **104**, such that the footwear **200** further encloses and compresses a user's foot in multiple directions for improved comfort and fit.

[0063] It is contemplated that the strap **208** is configured to automatically retract from the actuation configuration to the rest configuration. To that end, a biasing mechanism (not shown) may be included to return the strap **208** from the actuation configuration to the rest configuration, thereby decreasing the distance G between the midsection **236** of the strap **208** and the upper **102**, such that the strap **208** translates forward and downward relative to the longitudinal axis L (see FIG. 4). It is contemplated that the first segment **252** of the cord **212** may be formed of a flexible or elastic material that acts as a biasing mechanism to allow for deformation, i.e., stretching, when the tensional force is applied, and then to return to an undeformed state when the tensional force is released, such that the strap **208** is translated from the actuation configuration to the rest

configuration. In other embodiments, where the first segment **252** of the cord **212** may be inflexible or only partially flexible, the biasing mechanism (not shown) may be coupled to the first segment **252** at an end opposite the strap **208**, such that the biasing mechanism translates the first segment **252** of the cord **212** and the strap **208** to the rest configuration simultaneously. It is contemplated that the biasing mechanism (not shown) may be a spring, such as a coil spring, which is located within the disc **220** and operably connected to the first segment **252** of the cord **212**, although other configurations are possible.

[0064] Referring to FIGS. **4** and **5**, the footwear **200** is adjusted from a loosened configuration to a tightened configuration by pulling the actuation mechanism, i.e., the strap **208** to actuate the closure system **204**. Further, the footwear is adjusted from the tightened configuration to the loosened configuration by manipulating, e.g., twisting or pressing, the release mechanism, i.e., the release mechanism **216**, to unlock the closure system **204**. Accordingly, the closure system **204** is provided to allow a user to tighten and loosen the footwear **200** quickly and efficiently by providing one-touch actuation via the strap **208** and one-touch release via the release mechanism **216**. To that end, the closure system **204** is also configured to allow for re-tightening and/or increased tightening of the footwear **200** when desired, even after the footwear **200** is already in the tightened configuration. It is contemplated that varying levels of tightness may be provided for different activities. For example, more vigorous activities, such as sporting activities where rapid directional changes occur, or activities involving a higher degree of footwear reinforcement, such as training or hiking on uneven terrain, may require higher levels of tightness to prevent displacement of the foot within and/or from the interior cavity **106** of the upper **102**. As such, the closure system **204** is configured to provide varying levels of tightness when in the tightened configuration.

[0065] To that end, the tightened configuration can include a plurality of tightness levels for selection by a user. For example, an initial or first tightness level is reached upon a first actuation of the strap **208**, i.e., pulling the strap **208** a single time to reach the actuation configuration and allowing the strap **208** to return to the rest configuration. Then, a second tightness level is reached after a second actuation of the strap **208**, and a third tightness level is reached after a third actuation of the strap **208**, and so on. It is contemplated that several tightness levels may be provided by the closure system **204** and that the strap **208** may be iteratively actuated in this manner to reach each successive, incremental tightness level. It is also contemplated that higher levels of tightness may be reached by applying greater tensional force to the strap **208**, thereby pulling the strap **208** a greater distance *G* from the upper **102**. In this way, a user can select the tightness level of the footwear **200** best suited for the desired activity.

[0066] Referring to FIG. **6**, which is a schematic representation of a top plan view of the article of footwear **200**, the release mechanism **216** is illustrated as having a generally circular face **280** and located approximately centrally between the lateral and medial sides **116**, **118**, so as to be intersected by the longitudinal plane extending through the longitudinal axis *L*. Further, the insole **126** is illustrated within the interior cavity **106** of the upper **102**, and visible through the opening **166** in the heel region **112** of the upper **102**. In the illustrated embodiment of FIG. **6**, the strap **208** is in the rest configuration at the heel end **146** of the footwear **200**. In addition, the second segment **256** of the cord **212** is illustrated as extending across the upper **102** on the lateral and medial sides **116**, **118** within the midfoot and forefoot regions **110**, **108**. A portion of the second segment **256** extends through the retention member **260** in the top section **272** of the upper **102**.

[0067] For illustrative purposes, a yellow directional arrow is provided near the release mechanism **216** to indicate that rotation, i.e., twisting, the release mechanism **216** in the counterclockwise direction releases the closure system **204** from the tightened configuration. As a result, the cord **212** is released by the locking mechanism, e.g., the disc **220**, to allow the footwear **200** to be loosened. It is contemplated that the release mechanism **216** will also automatically return to its original position by rotating clockwise, such that the user does not need to hold the release mechanism **216**

in its rotated position to release the closure system **204** or loosen the footwear **200**. Such automatic retraction may be enabled by the same biasing mechanism (not shown) that retracts the strap **208**, or by a different biasing mechanism dedicated to the release mechanism **216** alone. Further, it is contemplated that the cord **212**, upon being released, may relax from a deformed or stretched state to an undeformed and relaxed state. In doing so, the compression force distributed along the footwear **200** and, particularly, on the user's foot is at least partially released. However, due to frictional forces that exist between the cord **212** and the upper **102** and/or sole **104**, such as within the channels (not shown) or by the retention member **260**, a partial amount of compression forces and tensional forces are still present immediately after the release mechanism **216** releases or unlocks the closure system **204**. Therefore, when in the released configuration, as depicted in FIG. **6**, the user can further expand and loosen the footwear **200** via contact with a portion of the footwear **200**, such as the heel region **112**, to pry the foot from the interior cavity **106** of the upper **102** without resistance by the cord **212**. In doing so, the cord **212** further loosens and relaxes within the upper **102** and/or the sole **104** to facilitate removal of the foot from the upper **102**.

[0068] FIG. **7** depicts another schematic representation of a lateral side **116** of an article of footwear **300** that is similar to the article of footwear **200**. In the illustrated embodiment, the footwear **300** has a closure system **304** that is similar to the closure system **204**. In particular, the closure system **304** includes an actuation mechanism in the form of a strap **308** that is operably engaged with a cord **312** to tighten and/or loosen the footwear **300**. The closure system **304** further includes a release mechanism in the form of a button **316** that is also operatively coupled to the cord **312** to loosen the footwear **300**. Further, the closure system **304** has a closure mechanism or disc **320** that is operably coupled to the strap **308** via the cord **312** for actuation to tighten and/or loosen the footwear **300**. The disc **320** is also operably connected to the button **316** for unlocking the closure system **304** and loosening the footwear **300**. In the illustrated embodiment, the strap **308** is arranged in the midfoot region **110** of the upper **102**. Further, the strap **308** is located on a top section **324** of the upper **102** and disposed centrally between the lateral side **116** and the medial side **118** of the footwear **300**. Accordingly, the strap **308** has a midsection **328** extending between first and second ends **332**, **336** that are positioned adjacent apertures **340a**, **340b** formed within the top section **324** of the upper **102**. It is contemplated that the apertures **340a**, **340b** are connected to channels (not shown) that extend into and along the upper **102** as a passage for the cord **312** and the strap **308**.

[0069] As illustrated in FIG. **7**, the button **316** and the disc **320** are arranged on the lateral side **116** of the upper **102** within the midfoot region **110**. In the illustrated embodiment, the button **316** and disc **320** are positioned adjacent the strap **308** and on a portion of the top section **324** of the upper **102**. The button **316** is configured to be pressed or pushed inwardly, i.e., toward the interior cavity **106**, into the upper **102** to release the closure system **304** from a locked or tightened configuration. In addition, the disc **320** circumscribes the button **316** and, thus, the disc **320** is also arranged on the lateral side **116** of the upper **102**.

[0070] As described in connection with the closure system **204**, the cord **312** includes a first segment **344** that extends along the upper **102** from the disc **320** and/or button **316** to the strap **308** and toward the medial side **118** of the footwear **300**. The first segment **344** of the cord **312** is operatively coupled to the strap **308** and the disc **320** and/or button **316** to actuate the closure system **304** to tighten the footwear **300**. Further, the cord **312** includes a second segment **348** that extends along the upper **102** from the disc **320** and/or button **316** and down the lateral side **116** of the footwear **300**. The second segment **348** of the cord **312** is operatively coupled to the button **316** and the disc **320**, and the second segment **348** extends along the upper **102** within the forefoot region **108** and midfoot region **110**. Both the first segment **344** and the second segment **348** of the cord **312** extend in a V-shaped pattern on respective medial and lateral sides **118**, **116**, such that each extend diagonally downwardly from the top section **324** toward the sole **104** before extending diagonally upwardly and back toward the top section **324** within the forefoot region **108** where the

first segment **344** crosses over the second segment **348** and, finally, terminate at anchors **352**, one of which is visible in FIG. 7. It is contemplated that the first segment **344** and the second segment **348** may be arranged on the footwear **300** in a different configuration, such that the second segment **348** crosses over the first segment **344** within the forefoot region **108**, or in another region of the footwear **300**, among other arrangements.

[0071] Accordingly, when the closure system **304** is actuated by pulling strap **308**, the first and second segments **344**, **348** are applied with a tensional force that is transmitted therealong to the anchors **352**. As a result, the first and second segments **344**, **348** are tightened between the disc **320** and the anchors **352**, which causes a compression force to be distributed on the upper **102** and/or the sole **104** to tighten the footwear **300** around a user's foot. In addition, the upper **102** of the footwear **300** includes a panel **354** that is superimposed thereon along the lateral side **116** and the medial side **118** and within the forefoot region **108**, the midfoot region **110**, and the heel region **112**. In the illustrated embodiment, the panel **354** is defined between a pair of edges **356** that interface with the upper **102** and extend in a curved manner, including various convex and concave curvatures therealong, across the footwear **300**. It is contemplated that one or both of the edges **356** may also include a third segment **358** of the cord **312** that extends therealong, such that the third segment **358** is operatively coupled to the disc **320** to further distribute the compression force along the upper **102** and/or the sole **104**. Further, the panel **354** includes holes or eyelets **360** through which the cord **312** is threaded or laced.

[0072] FIG. 8 depicts a schematic representation of the lateral side **116** of an article of footwear **400** similar to the footwear **300** and the footwear **200**. The footwear **400** has a closure system **404** that includes an actuation mechanism in the form of a strap **408**, a release mechanism **412** in the form of a switch or button, and a cord **416**. Further, the closure system **404** has a closure mechanism or disc **420** that is connected to the release mechanism **412** and operatively coupled to the cord **416** and the strap **408**, as illustrated in FIGS. 8 and 9. In the illustrated embodiment, the strap **408** extends from apertures **424a**, **424b** formed in a circular face **428** of the closure system **404** and a rotational axis **R** extends centrally through the face **428**. In particular, the release mechanism **412** and the disc **420** are intersected by and axially aligned with the rotational axis **R**, although other configurations are possible. In the illustrated embodiment, the cord **416** includes a first segment **432** and a second segment **436** (see FIG. 9) that each extend outwardly from first and second passages **440**, **444**, respectively, of the disc **420** to tighten the upper **102** and/or sole **104** of the footwear **400**. Referring to FIG. 8 again, the closure system **404** is provided as a consolidated unit or package configured to be centralized and mounted on the footwear **400**. By contrast, the closure systems **204** and **304** are decentralized in that the actuation mechanism, i.e., the straps **208** and **308**, and the closure mechanism, i.e., the discs **220** and **320**, are spaced apart from one another on the footwear **200** and **300**, respectively. Further, as illustrated in FIG. 8, a ring **446** is disposed about, e.g., circumscribes, the disc **420** and, although not depicted in FIG. 9, the ring **446** conceals the passages **440**, **444** through which the first segment **432** and the second segment **436** of the cord **416** extend (see FIG. 9). In some embodiments, the ring **446** is configured to be rotated or twisted about the rotational axis **R** to release or unlock the closure system **404**, in addition to or alternatively to the release mechanism **412**.

[0073] As illustrated in FIG. 9, the centralized closure system **404** has a shaft **448** to which the release mechanism **412** is attached for rotational movement relative to the disc **420**. Further, the disc **420** comprises a housing **450** that has a plurality of teeth **452** arranged radially on an inside surface **456** of a sidewall **460** thereof for rotational engagement with a gear (not shown) provided on a spool **464** of the release mechanism **412**. A portion of the cord **416** is wound around the spool **464** of the release mechanism **412**. In some embodiments, the first segment **432** of the cord **416** is threaded through the first passage **440** and wound around the spool **464**. In other embodiments, the second segment **436** of the cord **416** is threaded through the second passage **444** and wound around the spool **464**. In still other embodiments, both the first and second segments **432**, **436** are wound

around the spool **464**. In addition, the disc **420** receives a coil spring **468** concentrically disposed within a chamber **472** of the housing **450** and axially aligned with the rotational axis R. When the closure system **404** is assembled, the coil spring **468** is positioned between the disc **420** and the spool **464** of the release mechanism **412**, although other configurations are possible.

[0074] In one implementation in which the second segment **436** is wound around the spool **464**, the closure system **404** is actuated by a user applying a tensional force, i.e., pulling, the strap **408** away from the release mechanism **412** and against the force of the coil spring **468** to reach an actuated configuration (as illustrated in FIG. 9), which causes the spool **464** of the release mechanism **412** to rotate about the rotational axis R. As the spool **464** rotates, the face **428** of the release mechanism **412** and the disc **420** both remain fixed in place and do not rotate about the rotational axis R. Further, as the spool **464** rotates, the second segment **436** of the cord **416** is retrieved through the second passage **444** and wound around the spool **464**, which causes the tensional force to be applied to the second segment **436** and conveyed to the footwear **400** as a compression force for tightening the footwear **400**. In addition, as the spool **464** is rotated about the rotational axis R within the disc **420**, the gear (not shown) engages with the plurality of teeth **452** arranged on the disc **420** to prevent backwards rotation or unwinding and to iteratively secure the spool **464** in its rotational displacement about the rotational axis R. Due to the mechanical interaction between the gear (not shown) and the plurality of teeth **452** of the disc **420**, audible sounds resembling “clicks” and perceptible vibrations may be produced for providing acoustic and haptic feedback to a user, such that the user can associate such feedback with operation of the closure system **404**.

[0075] Subsequently, after the user releases the strap **408**, the coil spring **468** retracts the strap **408** back toward the release mechanism **412** into the rest configuration, while the spool **464** remains locked in its radial displacement position by interaction between the gear (not shown) and the plurality of teeth **452** on the disc **420**. At this point, when the spool **464** is locked in its radial displacement position and the strap **408** has returned to the rest configuration, the first segment **432** and the second segment **436** are locked or secured against movement and, thus, the closure system **404** has reached a first tightening level of a tightening configuration of the footwear **400**. It is contemplated that pulling the strap **408** again will cause farther radial displacement of the spool **464** within the disc **420** to reach a second tightening level of increased compression force, i.e., tighter, than the first tightening level, at which point the spool **464** is locked into place by the interaction of the gear (not shown) and the plurality of teeth **452**. In this manner, increased tightening levels may be achieved beyond the second tightening level, as desired by a user. To release or unlock the closure system **404** from the tightened configuration, the user can press down on the face **428** of the release mechanism **412** to move the gear (not shown) out of engagement with the plurality of teeth **452**, thereby unlocking the spool **464** from the disc **420** and allowing reverse or backward rotation.

[0076] In another implementation, the closure system **404** is released or unlocked by rotating or twisting the ring **446** (see FIG. 8). Referring to FIGS. 8 and 9, the ring **446** can be configured to be rotated to disengage the gear (not shown) of the release mechanism **412** from the plurality of teeth **452** of the disc **420**. For example, the ring **446** may be coupled to the release mechanism **412** to cause axial displacement of the release mechanism **412** from the disc **420**, such that the gear (not shown) provided on the spool **464** of the release mechanism **412** becomes axially displaced out of engagement with the plurality of teeth **452** of the disc **420**. As a result, the first segment **432** and the second segment **436** of the cord **416** are unlocked or loosened and, thus, the footwear **400** is adjusted from the tightened configuration to the loosened configuration by rotation of the ring **446**.

[0077] FIG. 10 illustrates another schematic representation of lateral side view of an article of footwear **500** having a closure system **504** that is arranged within the midfoot **110** of the footwear **500**. It is contemplated that the closure system **504** is similar to the centralized, packaged closure system **404**. In the illustrated embodiment, the closure system **504** has a combined actuation mechanism and release mechanism in the form of a latch assembly **508** extending from the lateral

side **116** to the medial side **118** of the sole **104** and around the midfoot region **110** of the upper **102**. Further, the closure system **504** has a closure mechanism or disc **512** mounted on the sole **104** within an arch region **516** of the midfoot region **110**. The arch region **516** is a region corresponding to the location of the arch of a user's foot within the upper **102** when the footwear **500** is worn. The disc **512** includes external ridges **520** that are exposed and configured to be accessible to a user within the arch region **516**. In the illustrated embodiment, the latch assembly **508** includes a hook **524** that is positioned at a first distal end **528** of a first wireframe **532** that extends from a first base (not shown) disposed on the medial side **118** of the footwear **500**. As illustrated in FIG. **10**, the hook **524** is positioned in the rest configuration and configured to be actuated by articulation toward the lateral side **116** of the footwear **500**. In addition, the latch assembly **508** includes a latch **540** at a second distal end **542** on a second wireframe **544** that extends from a second base **548** disposed on the lateral side **116** of the footwear **500**. Similar to the hook **524**, the latch **540** is illustrated in the rest configuration in FIG. **10** and configured to be articulated toward the medial side **118** of the footwear **500**. The second wireframe **544** forms a slot **552** into which the hook **524** is received when the hook **524** is articulated, e.g., by bending or rotating the first wireframe **532** about the first base (not shown), toward the lateral side **116** and the latch **540**.

[0078] To loosen or unlock the closure system **504**, a user may unfasten the hook **524** from the latch **540** by articulating the hook **524** toward the lateral side **116** beyond engagement with the latch **540**, which may also be assisted by articulating the latch **540** toward the medial side **118**, and then pulling vertically upward, away from the upper **102**, until the hook **524** is retracted back toward the medial side **118** by the self-biasing forces provided by the first wireframe **532** and the first base (not shown) due to bending and/or displacement from the rest configuration. To that end, the first wireframe **532** and the second wireframe **544** are preferably formed of a resilient and flexible material. Further, the closure system **504** can be tightened or loosened by manipulation, e.g., rotation, of the disc **512** to translate the first base (not shown) and the second base **548** toward each other. Alternatively, operation of the disc **512** may retrieve portions of the first wireframe **532** and/or the second wireframe **544**, such that the latch assembly **508** becomes increasingly restricted around the upper **102** and, thus, the user's foot when the footwear **500** is worn. In some embodiments, the disc **512** can be rotated counterclockwise to tighten the footwear **500** and clockwise to unlock or release the footwear **500** from the tightened configuration, thereby allowing the user to loosen the footwear **500**.

[0079] Referring to FIG. **11**, which depicts a schematic representation of an article of footwear **600**, a closure system **604** is arranged on the sole **104** within the midfoot region **110** of the footwear **600**. In the illustrated embodiment, the closure system **604** includes a closure mechanism or **608** mounted to the bottom surface **176** of the sole **104** and operably engaged with a cord **612** for tightening and loosening the footwear **600**. It is contemplated that the closure system **604** can incorporate any of the actuation mechanisms and release mechanisms described in connection with the closure systems **204**, **304**, **404**, **504**. As illustrated in FIG. **11**, the cord **612** is operably engaged with the disc **608** and includes multiple segments **616a-h** extending across the sole **104** and the upper **102** in a web-like winding pattern. By locating the disc **608** on the sole **104** and within the midfoot region **110** of the footwear **600**, the mass of the disc **608** is centralized on the footwear **600** to provide a balanced distribution of mass or center of gravity of the footwear **600**, which may be desirable for applications involving high performance characteristics. In addition, the disc **608** is at least partially concealed underneath the footwear **600** so as to be inconspicuous, which may be desirable for aesthetic purposes.

[0080] FIG. **12** illustrates a schematic representation of a top plan view of a sole **104** of an article of footwear **700** having a closure system **704**. In the illustrated embodiment, the closure system **704** includes a closure mechanism or disc **708** mounted on the sole **104** in the midfoot region **110**. In particular, the disc **708** is at least partially embedded in a top surface **712** of the sole **104** and centrally disposed between the lateral side **116** and the medial side **118** of the footwear **700**.

Further, the disc **708** is located within an arch **716** of the sole **104**, which is a region corresponding to a location of the arch of a user's foot within the upper **102** when the footwear **700** is worn. It is contemplated that the closure system **704** can incorporate any of the actuation mechanisms and release mechanisms described in connection with the closure systems **204**, **304**, **404**, **504**. As illustrated in FIG. **12**, a cord **720** is operably engaged with the disc **708** and includes multiple segments **724a-h** extending across the sole **104** in a web-like winding pattern.

[0081] In some embodiments, the footwear **700** may include the insole **126** (see FIG. **2**) to cover the disc **708**, such that the disc **708** is encapsulated between the insole **126** and other portions of the sole **104** and accessible by removal or displacement of the insole **126**. In some embodiments, an access plug (not shown) may be provided on the sole **104** of the footwear **700** for accessing the disc **708**. Further, the disc **708** may be entirely enclosed and inaccessible without permanent deformation of the footwear **700**. Whether the disc **708** is accessible or inaccessible, encapsulating the disc **708** provides protection from damage due to impact with external surfaces or objects during use, from outdoor environments, or even damage due to manipulation or impact applied by the user. By locating the disc **708** within the sole **104** and within the midfoot region **110** of the footwear **700**, the mass of the disc **708** is centralized on the footwear **600** to provide a balanced distribution of mass or center of gravity of the footwear **700**, which may be desirable for applications involving high performance characteristics. In addition, the disc **708** is concealed so as to be inconspicuous, which may be desirable for aesthetic purposes. It is contemplated that the closure system **704** can be embedded within the sole **104**, so as to be encapsulated entirely within the sole **104**, or, it could be at least partially encapsulated or embedded within the sole **104** for protection from outdoor environments, collisions due to use, and the like.

[0082] It is also contemplated that any of the closure systems **204**, **304**, **404**, **504**, **604**, **704** described herein may incorporate a closure mechanism similar to those disclosed in U.S. Pat. Nos. 5,325,613, 5,600,875, 5,606,778, 5,638,588, 5,651,198, and 5,669,116, which are all commonly assigned to Puma SE and incorporated by reference in their entirety herein. For example, it is contemplated that the disc **220**, **320**, **420**, **512**, **608**, **708** may include one or more aspects of such closure mechanisms to provide tightening or loosening functionality when mounted on the respective footwear **200**, **300**, **400**, **500**, **600**, **700** of the present disclosure.

[0083] In other embodiments, other configurations are possible. For example, certain features and combinations of features that are presented with respect to particular embodiments in the discussion above can be utilized in other embodiments and in other combinations, as appropriate. Further, any of the embodiments described herein may be modified to include any of the structures or methodologies disclosed in connection with other embodiments. Additionally, the present disclosure is not limited to articles of footwear of the type specifically shown. Still further, aspects of the articles of footwear of any of the embodiments disclosed herein may be modified to work with any type of footwear, apparel, or other athletic equipment.

[0084] As noted previously, it will be appreciated by those skilled in the art that while the invention has been described above in connection with particular embodiments and examples, the invention is not necessarily so limited, and that numerous other embodiments, examples, uses, modifications, and departures from the embodiments, examples, and uses are intended to be encompassed by the claims attached hereto. The entire disclosure of each patent and publication cited herein is incorporated by reference, as if each such patent or publication were individually incorporated by reference herein. Various features and advantages of the invention are set forth in the following claims.

INDUSTRIAL APPLICABILITY

[0085] Numerous modifications to the present invention will be apparent to those skilled in the art in view of the foregoing description. Accordingly, this description is to be construed as illustrative only and is presented for the purpose of enabling those skilled in the art to make and use the

invention. The exclusive rights to all modifications which come within the scope of the appended claims are reserved.

Claims

1. An article of footwear, comprising: an upper that is attached to a sole; a longitudinal axis that extends between a rear end and a front end of the article of footwear; a midfoot region that is disposed between a forefoot region and a heel region; a heel component disposed adjacent the rear end within the heel region, the heel component having a midsection that includes padding; a fastening system including a fastening mechanism that is operably engaged with a cord; and a panel, wherein the panel includes an edge, wherein the edge is curved, wherein the cord is laced through a plurality of eyelets arranged on the panel, wherein the cord extends on a lateral side of the upper so that a first segment extends diagonally downward from a top section toward the sole before extending diagonally upwardly toward the top section, and wherein the first segment crosses over a second segment of the cord.
2. The article of footwear of claim 1, wherein the fastening mechanism is configured to be twisted or rotated by a user.
3. The article of footwear of claim 1, wherein the cord extends in a V-shaped pattern on the lateral side of the upper and a lowermost point of the V-shaped pattern of the cord is positioned on the upper in the midfoot region.
4. The article of footwear of claim 1, wherein the fastening mechanism is configured to be iteratively actuated to reach successive tightness levels.
5. The article of footwear of claim 1, wherein the fastening mechanism is arranged on the lateral side of the upper.
6. The article of footwear of claim 1, wherein the fastening mechanism is configured to be pushed inwardly to release the fastening mechanism from a tightened configuration.
7. The article of footwear of claim 1, wherein the panel is superimposed on the lateral side and a medial side of the upper within the forefoot region and the midfoot region.
8. The article of footwear of claim 1, wherein the cord extends in a V-shaped pattern.
9. The article of footwear of claim 1, wherein upper is formed of a hydrophobic material.
10. An article of footwear, comprising: an upper that is attached to a sole; a longitudinal axis that extends between a rear end and a front end of the article of footwear; a midfoot region that is disposed between a forefoot region and a heel region; a heel component disposed adjacent the rear end within the heel region, the heel component having a midsection that includes padding; a fastening system including a closure mechanism that is operably engaged with a cord; and a panel, wherein a distance is defined between the midsection of the heel component and the upper, wherein the distance is configured to be adjusted by a force applied by a user between a first configuration and a second configuration, wherein, in the first configuration, the midsection of the heel component is configured to be translated rearward relative to the longitudinal axis, and wherein, in the second configuration, the midsection of the heel component is configured to be translated forward relative to the longitudinal axis, wherein the panel includes a first edge, wherein the first edge is curved, and wherein the cord is routed through the panel.
11. The article of footwear of claim 10, wherein the midsection of the heel component curves concavely between a lateral side and a medial side relative to a toe end.
12. The article of footwear of claim 10, wherein the cord defines a winding region that extends across the upper.
13. The article of footwear of claim 10, wherein the panel includes a second edge that is opposite the first edge, wherein the second edge includes convex curvatures therealong.
14. The article of footwear of claim 10, wherein the upper is formed of a hydrophobic material.
15. The article of footwear of claim 10, wherein the heel component is configured to retain its

shape in a rest configuration.

16. An article of footwear, comprising: an upper that is attached to a sole and includes a panel defined between a pair of edges; a longitudinal axis that extends between a heel end and a toe end of the article of footwear; a midfoot region that is disposed between a forefoot region and a heel region; a fastening system arranged on the upper and including a cord; and a heel component disposed adjacent the heel end of the upper within the heel region and extending from the heel end of the upper in a rearward direction relative to the longitudinal axis, wherein the heel component includes a midsection that is relatively thicker than surrounding portions of the heel component, and wherein the midsection curves concavely between a lateral side and a medial side of the heel component relative to the toe end of the article of footwear, and wherein the pair of edges of the panel are curved.

17. The article of footwear of claim 16, wherein the cord extends in a V-shaped pattern on a lateral side of the upper, such that the cord extends diagonally downwardly from a top section toward the sole before extending diagonally upwardly toward the top section.

18. The article of footwear of claim 16, wherein a plurality of retention members are arranged on the upper for retaining the cord.

19. The article of footwear of claim 16, wherein the fastening system includes a closure mechanism and a release mechanism.

20. The article of footwear of claim 19, wherein the closure mechanism is configured to be twisted by a user and wherein the release mechanism is configured to be pressed by a user.
